UNIT 2 · TOPIC 4(A)

Dynamic Balancing of Farm Machines

Syllabus topic: Dynamic balancing and stability of farm machines

Static vs dynamic balance

	Condition
Static balance	Resultant centrifugal force of all rotating masses = 0 (CG on the axis of rotation). A statically balanced rotor rests in any angular position.
Dynamic balance	Resultant force AND resultant couple both zero. Masses in different planes can be statically balanced yet dynamically unbalanced (rocking couple) — balance them in two correction planes.

Farm relevance: balance thresher cylinders, chaff-cutter flywheels, rotavator rotors and combine cylinders dynamically at the works (high speed + operator safety). An unbalanced cylinder causes vibration, bearing failure and frame cracking.

Rotating unbalance

$$F = m{\cdot}r{\cdot}\omega^2 \text{ — grows with the square of speed (doubling rpm quadruples the force)}$$

Case	How to balance
Single plane (static)	Place a balance mass m_b·r_b opposite the unbalance.
Several masses, one plane	Close the force polygon (vector sum m·r = 0).
Masses in several planes	Transfer them to two reference planes; close the force and couple polygons (two-plane/dynamic balancing machines do this).

Reciprocating unbalance (Kepner §14.11 — mower example)

$$F_h \approx [(W_s + W_2)/g]{\cdot}R{\cdot}\omega^2{\cdot}(\cos\varphi - K{\cdot}\cos 2\varphi + \text{offset term}),\ K = R/\sqrt{(L^2 - S^2)}$$

- This crank–slider inertia force (Kepner eq. 14.1) is a dominant **primary** force (frequency = crank speed) plus a smaller **secondary** force (2× crank frequency). The approximation holds when **K < 0.25**.
- A rotating counterweight **cannot** balance a reciprocating force: its centrifugal force stays constant in magnitude at all crank angles, while the reciprocating force varies with crank angle.
- Split the pitman weight into two concentrated parts: $\mathbf{W_1}$ at the crankpin (rotating) and $\mathbf{W_2}$ at the knife head (reciprocating) — the standard connecting-rod idealization.

Practical compromise — "half balancing": size the rotating counterweight opposite the crankpin to balance the purely rotating part (crankpin + rotating share of the pitman, F_c) plus about half the maximum reciprocating force:

$$\text{Counterweight force} = F_c + \tfrac{1}{2}F_r,\text{max}$$

This minimizes, but does not remove, the unbalance; the residual appears at right angles.

Complete balancing options (Kepner §14.12)

- **A second reciprocating counterweight** moving in direct opposition (dynamically balanced mower — counterweight on the inner shoe). The opposed masses must be collinear, or they create a vibratory couple.
- **Two opposed reciprocating knives** (double-knife cutter bar): forces cancel; each knife needs only half the stroke, allowing nearly double crank speed and higher forward speed.
- **Scotch-yoke double-knife drives** run at **1800–2000 rpm** in an oil bath.
- **Baler plunger (65–90 strokes/min):** a heavy flywheel + crank drive absorbs the large cyclic load; the flywheel supplies energy during the compression stroke and recovers between strokes.

Flywheels and speed fluctuation

$$E = \tfrac{1}{2} I\omega^2$$

$$C_s = (\omega_{max} - \omega_{min})/\omega_{mean};\ I = \Delta E/(C_s \cdot \omega^2_{mean})$$

- A flywheel stores kinetic energy and smooths cyclic torque demand (chaff cutter, baler, thresher drive). ΔE = fluctuation of energy per cycle.
- On threshers and chaff cutters the cutting/threshing cylinder itself acts as a flywheel.

Vibration basics

$$\omega_n = \sqrt{(k/m)}$$

- **Resonance:** when the excitation frequency ≈ the natural frequency, amplitudes become large. Avoid running rotors near frame natural frequencies.
- **Isolation:** low-stiffness mounts give transmissibility < 1 when the frequency ratio is **> √2**.
- Operator vibration limits and effects: Unit 5 (ergonomics).

Practice MCQs

Q46. A rotor at rest in any angular position without turning is:

(a) Dynamically balanced
(b) Statically balanced
(c) Unbalanced
(d) Over-balanced

Q47. Static balance requires:

(a) ΣForces = 0
(b) ΣCouples = 0
(c) Both zero
(d) High speed

Q48. Dynamic balance additionally requires:

(a) Zero resultant couple
(b) Higher mass
(c) One correction plane
(d) Rigid shaft

Q49. Masses balanced in one plane but distributed along the shaft may still produce:

(a) No effect
(b) A rocking (unbalanced) couple
(c) Static unbalance
(d) Torque rise

Q50. Centrifugal force due to unbalance varies as:

(a) ω
(b) ω²
(c) ω³
(d) √ω

Q51. Doubling the speed of an unbalanced thresher cylinder increases the unbalance force:

(a) 2×
(b) 4×
(c) 8×
(d) No change

Q52. Complete dynamic balancing of rotating masses in several planes needs:

(a) One counterweight
(b) Correction in two planes
(c) Flywheel
(d) Damper

Q53. A reciprocating inertia force CANNOT be fully balanced by a rotating counterweight because:

(a) It is too large
(b) Rotating mass gives constant-magnitude force while reciprocating force varies with crank angle
(c) Speeds differ
(d) Friction interferes

Q54. In mower balancing practice, the rotating counterweight is sized to balance the rotating parts plus about:

(a) One-fourth
(b) One-half of maximum reciprocating force
(c) Full reciprocating force
(d) Twice the reciprocating force

Q55. In the analysis of mower pitman forces, the pitman mass is idealized as:

(a) Uniform rod
(b) Two concentrated masses at crankpin (rotating) and knife head (reciprocating)
(c) Point at centre
(d) Neglected

Q56. The primary reciprocating inertia force has a frequency equal to:

(a) Half crank speed
(b) Crank speed
(c) Twice crank speed
(d) Four times

Q57. Two opposed reciprocating counter-masses fully cancel forces only if they are:

(a) Unequal
(b) Collinear (otherwise a vibratory couple results)
(c) At right angles
(d) On separate shafts

Q58. The double-knife (opposed knives) cutter bar permits:

(a) Lower cutting speed
(b) Half stroke per knife and higher crank rpm → higher forward speed
(c) Heavier knives
(d) No guards in all versions

Q59. In a dynamically balanced mower (Kepner Fig. 14.6), the balancing element is:

(a) Heavier flywheel
(b) Reciprocating counterweight on the inner shoe
(c) Rubber mounts
(d) Longer pitman

Q60. The energy-smoothing member for the baler plunger's cyclic load is the:

(a) Knotter
(b) Flywheel
(c) Pickup
(d) Bale tension rail

Q61. Coefficient of fluctuation of speed is:

(a) $(\omega_{max}-\omega_{min})/\omega_{mean}$
(b) $\omega_{mean}/(\omega_{max}-\omega_{min})$
(c) $\omega_{max}/\omega_{min}$
(d) $\Delta\omega^2$

Q62. Required flywheel moment of inertia for energy fluctuation ΔE is:

(a) $\Delta E \cdot C_s \cdot \omega^2$
(b) $\Delta E/(C_s \cdot \omega^2_{mean})$
(c) $C_s/\Delta E$
(d) $\frac{1}{2}I\omega^2$

Q63. Resonance occurs when:

(a) Damping is high
(b) Excitation frequency equals natural frequency
(c) Mass is large
(d) Speed is low

Q64. Natural frequency of a spring-mass system is:

(a) √(m/k) (b) √(k/m)

(c) k/m (d) m/k

Q65. Vibration isolation (transmissibility <1) requires frequency ratio greater than:

(a) 1 (b) √2

(c) 2 (d) π

Q73. An unbalanced high-speed cylinder in a thresher primarily causes:

(a) Better threshing (b) Vibration, bearing failure and frame cracks

(c) Higher capacity (d) Less noise

Answers

Q46 — (b) Statically balanced

Q47 — (a) ΣForces = 0

Q48 — (a) Zero resultant couple

Q49 — (b) A rocking (unbalanced) couple

Q50 — (b) ω²

Q51 — (b) 4× — *$F = mr\omega^2 \rightarrow 2^2 = 4\times$.*

Q52 — (b) Correction in two planes

Q53 — (b) Rotating mass gives constant-magnitude force while reciprocating force varies with crank angle — *Kepner §14.11 — the rotating counterweight force is constant at all crank angles.*

Q54 — (b) One-half of maximum reciprocating force — *Kepner §14.11 — compromise counterweight = F_c + ½$F_{r,max}$ ("half balancing"), residual unbalance shifted 90°.*

Q55 — (b) Two concentrated masses at crankpin (rotating) and knife head (reciprocating)

Q56 — (b) Crank speed

Q57 — (b) Collinear (otherwise a vibratory couple results)

Q58 — (b) Half stroke per knife and higher crank rpm → higher forward speed — *Kepner §14.12 — each knife stroke halves; crank speed (and forward speed) can nearly double for the same inertia stress.*

Q59 — (b) Reciprocating counterweight on the inner shoe

Q60 — (b) Flywheel

Q61 — (a) $(\omega_{max}-\omega_{min})/\omega_{mean}$

Q62 — (b) $\Delta E/(C_s \cdot \omega^2_{mean})$

Q63 — (b) Excitation frequency equals natural frequency

Q64 — (b) √(k/m)

Q65 — (b) √2 — *Transmissibility < 1 only when $\omega/\omega_n > \sqrt{2}$ — basis of soft engine mounts.*

Q73 — (b) Vibration, bearing failure and frame cracks

Sources: Kepner, Bainer & Barger — Principles of Farm Machinery, Ch. 1 (Research & Development), Ch. 2 (Field Capacities), §14.11 (Reciprocating Unbalance in a Mower); theory-of-machines standards (balancing, flywheels); tractor–implement mechanics references.